

TITLE & AUTHOR INFORMATION

Air Particulate Matter Exposure is Associated with Non-Shockable Cardiac Rhythm in Out-of-Hospital Cardiac Arrest: A Community-Based Analysis of Alachua County

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STUDY OVERVIEW

BACKGROUND

- Death and disability from out-of-hospital cardiac arrest (OHCA) remains a prominent public health threat in the United States
- Air particulate matter less than 2.5 microns (PM2.5) is an independent risk factor for cardiovascular mortality and morbidity
- Previous studies have reported the association between short-term PM2.5 exposure and the risk of out-of-hospital cardiac arrest, however its relationship to the cardiac rhythm at OHCA onset is unknown.

OBJECTIVE

- We investigated the relationship between location-controlled, short-term and acute PM2.5 exposure and initial cardiac rhythm in OHCA

STUDY DESIGN

- In this retrospective cohort study, emergency medical service records of adult, non-traumatic OHCA within Alachua County, FL from 2012-2014 were reviewed and matched with UF Health electronic records
- Patients who did not survive to the emergency department or unable to be geocoded were excluded to avoid variable misclassification (**Figure 1**)
- Locations of cardiac arrest incidence were geocoded and spatially joined to corresponding census tract (ArcGIS) (**Figure 2**)
- Daily, census tract-level PM2.5 measurements 0, 1, 2, and 3 days before event were assigned to each subject

Primary Exposure

Daily, census tract-level fine particulate matter (PM2.5)

CDC Public Health Tracking Network

Primary Outcome

Presenting cardiac rhythm determined by EMS on arrival

Non-Shockable vs. Shockable Rhythms

STATISTICAL ANALYSIS

- Descriptive statistics were applied to assess cohort by outcome (**Table 1**)
- Wilcoxon rank-sum test was used to compare exposure by outcome (**Figure 3**)
- Multilevel mixed-effects logistic regression model was developed to assess the relationship between PM2.5 exposure and the probability of non-shockable initial cardiac rhythm (**Figure 4, Figure 5**)
- Intraclass correlation coefficient (ICC) was evaluated to assess the geographic and temporal variation of exposure/outcome (**Table 2**)

CONCLUSION

- Short-term, location-controlled PM2.5 exposure is associated with a less desirable initial cardiac rhythm in OHCA
- This research suggests that short-term fine particulate matter exposure may be an indicator or modifier of OHCA etiology.

In Alachua County, short-term air pollution exposure increases odds of non-shockable cardiac rhythm during out-of-hospital cardiac arrest



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TABLES & FIGURES

Figure 1: Study inclusion flow diagram

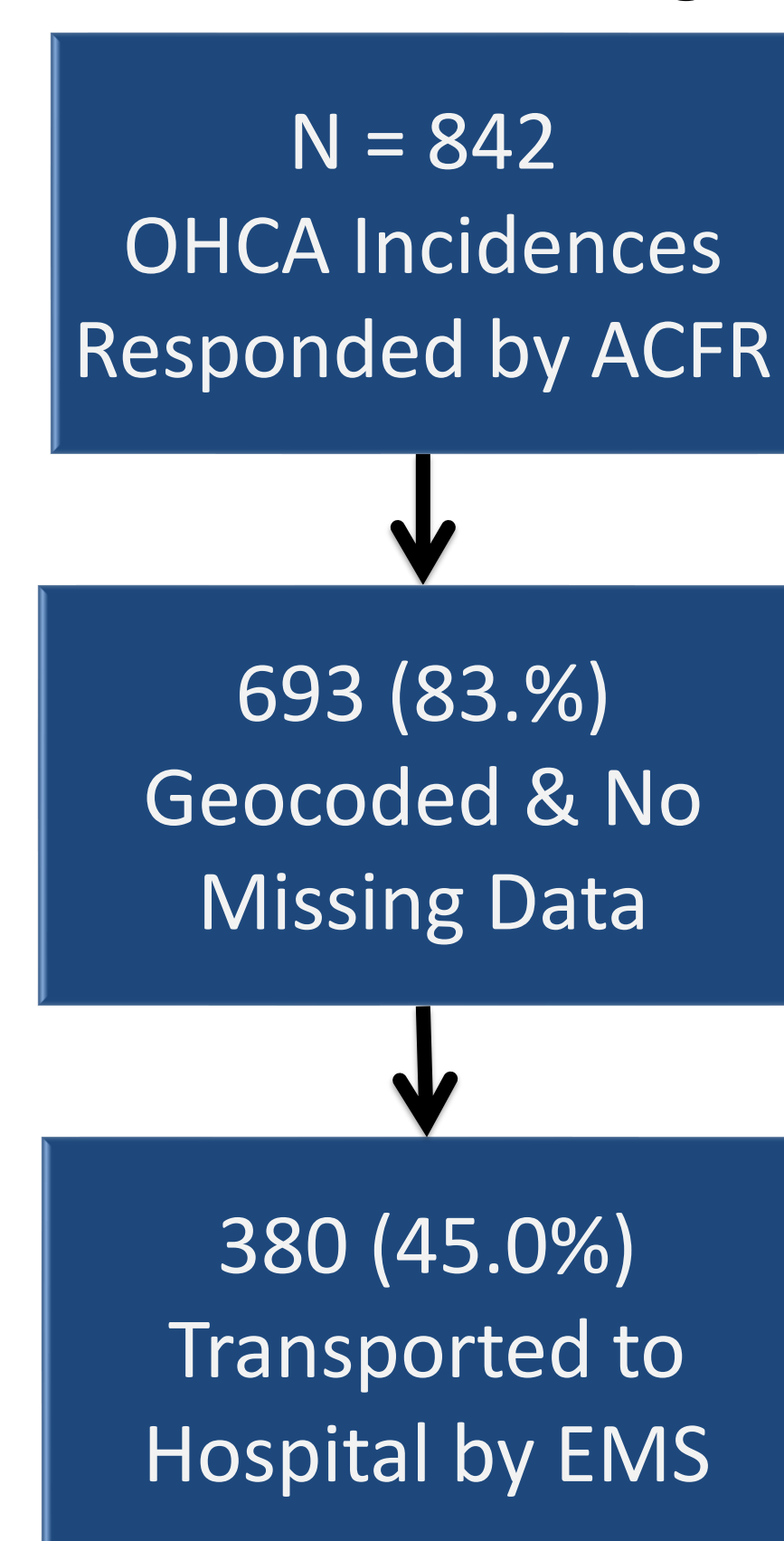


Figure 2: Heat map of population-adjusted spatial density of OHCA incidence (OHCA/km²/year) [Range: 0 – 60]

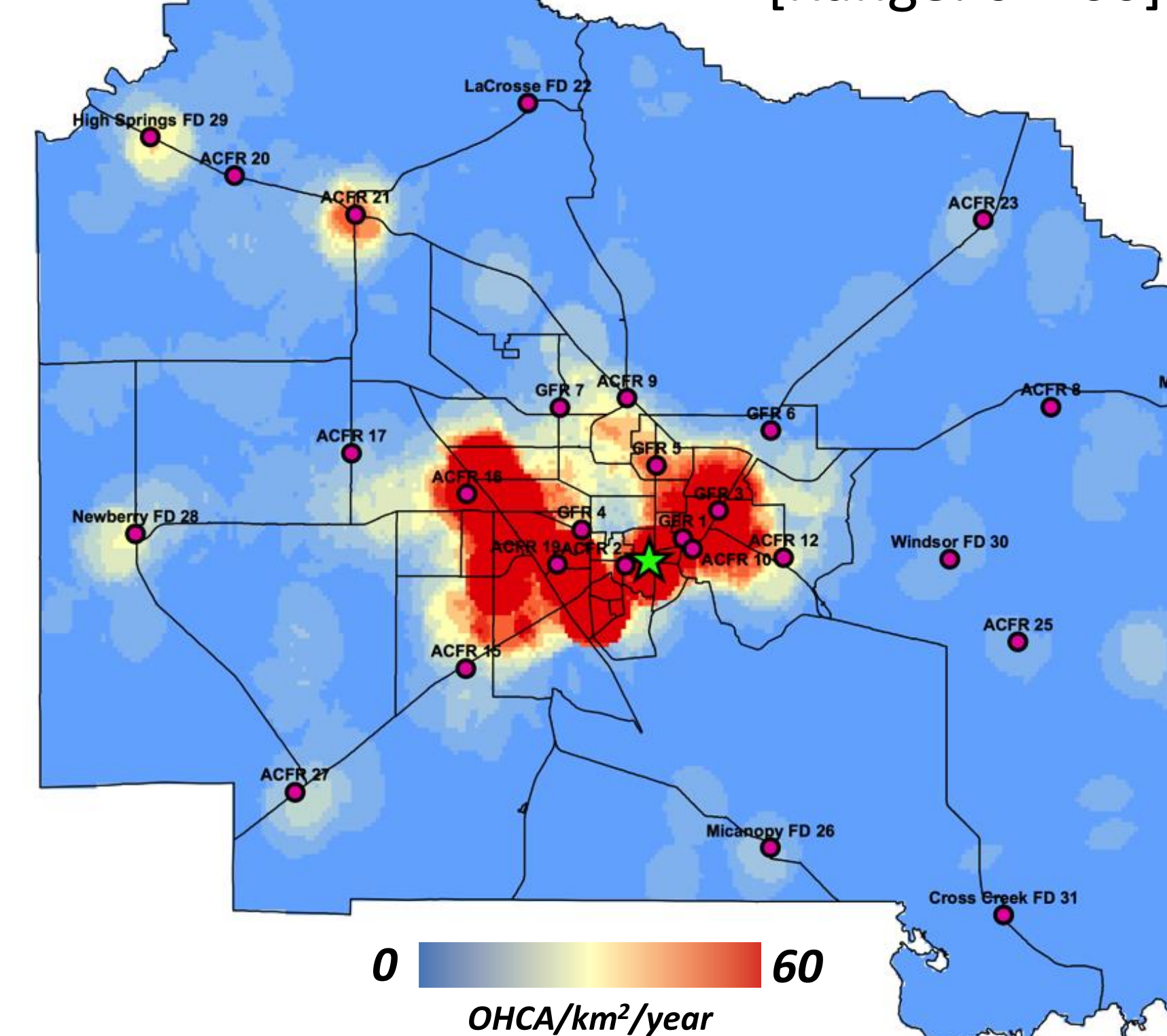


Figure 3: Particulate matter exposure by initial cardiac rhythm on day of event

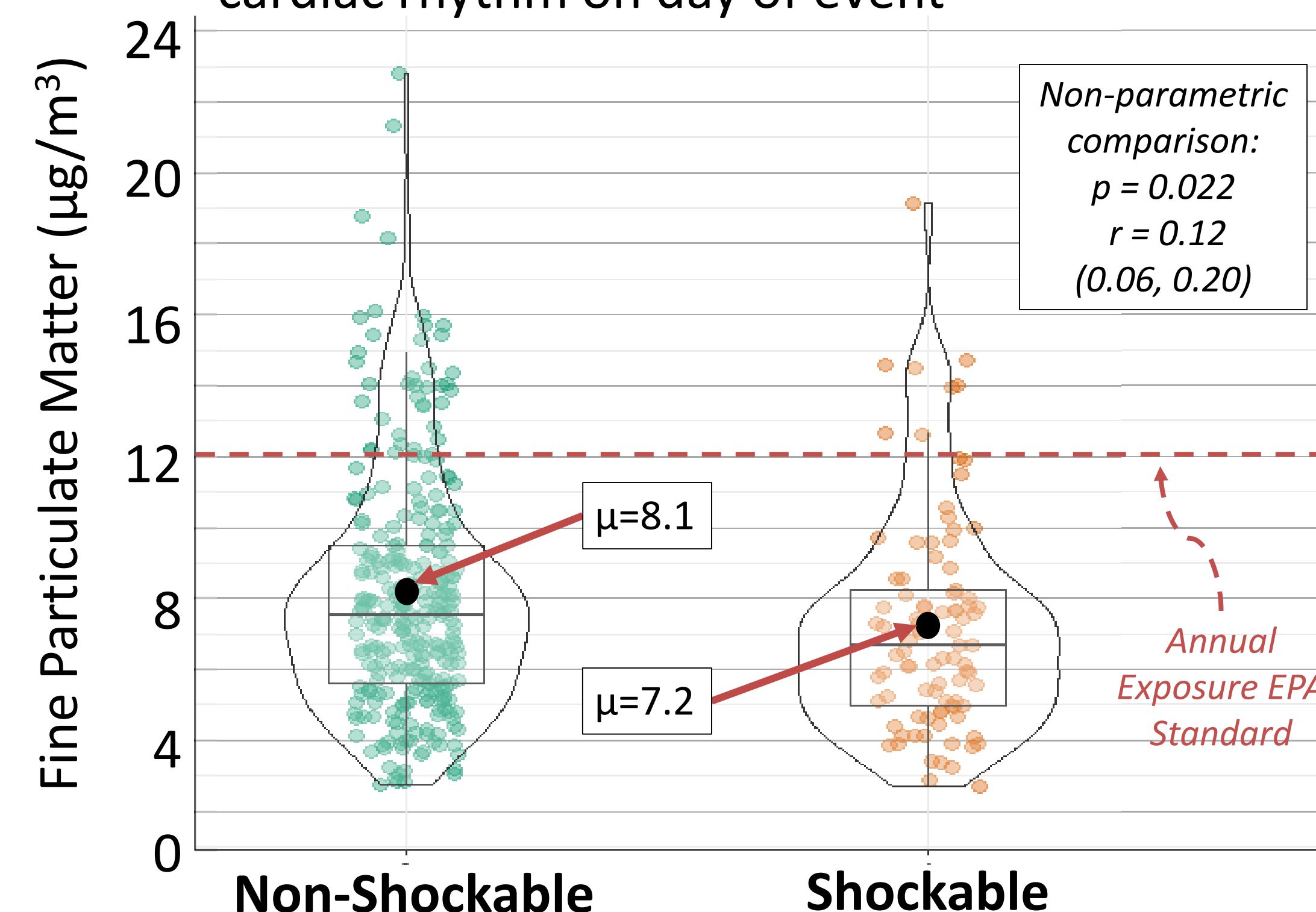


Table 2: Assessment of geographic and temporal variation in exposure/outcome using multilevel models

Cluster Variation ICC (P-value)	Particulate Matter Exposure	Non-Shockable Probability
Geography	2.2% (0.229)	11.3% (0.036)
Day of Week of Event	2.0% (0.171)	0.0% (0.999)
Week of Event	17.9% (0.002)	0.0% (0.999)
Month of Event	12.6% (0.022)	0.0% (0.999)
Season of Event	8.3% (0.103)	0.0% (0.999)

Table 1: Study demographics, predictors, and outcomes of eligible OHCA cohort by initial cardiac rhythm

Patient Predictors	Non-Shockable (n=287, 75.5%)	Shockable (n=93, 24.5%)	P-value
Age, mean (SD)	64.9 (17.4)	66.0 (16.2)	0.574
Sex (% Female)	136 (47.4)	27 (29.0)	0.003
Race:			
Black/African-Amer.	102 (35.5)	35 (26.2)	0.229
White	184 (64.1)	96 (72.4)	
Other/Unknown	1 (0.4)	18 (1.4)	
Bystander AED:	8 (4.3)	6 (11.5)	0.088
Bystander CPR:	91 (49.2)	33 (63.5)	0.084
Time of Day (AM vs. PM)	113 (39.4)	43 (41.9)	0.715
Survival to Discharge*	14 (10.5)	18 (40.9)	<0.001
Survival to Home Discharge	9 (6.7)	10 (22.7)	0.008

Figure 4: Odds ratios (OR) for having non-shockable rhythm at every 50% increase in daily PM2.5 exposure

Number of Days Before Event (Lag Day)				
	Lag 0	Lag 1	Lag 2	Lag 3
OR	1.29	1.09	1.13	1.12
P-value	0.030	0.442	0.309	0.314

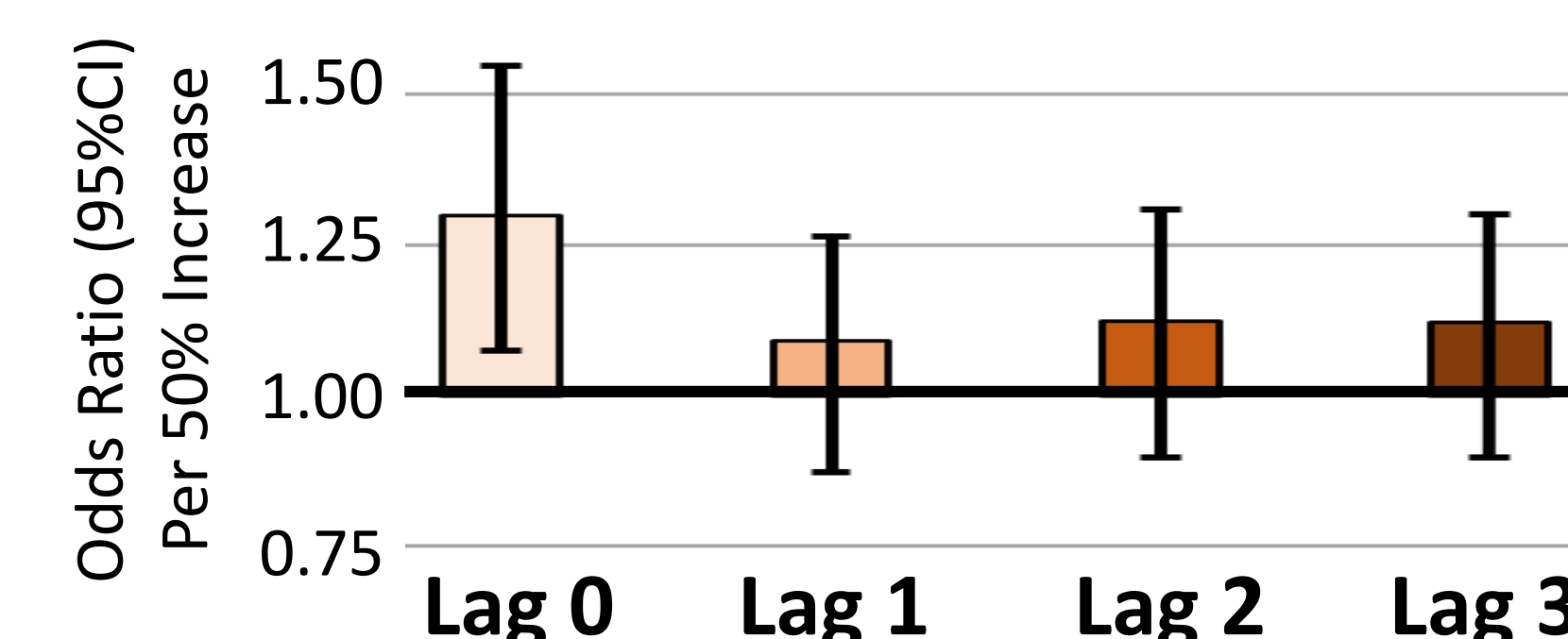
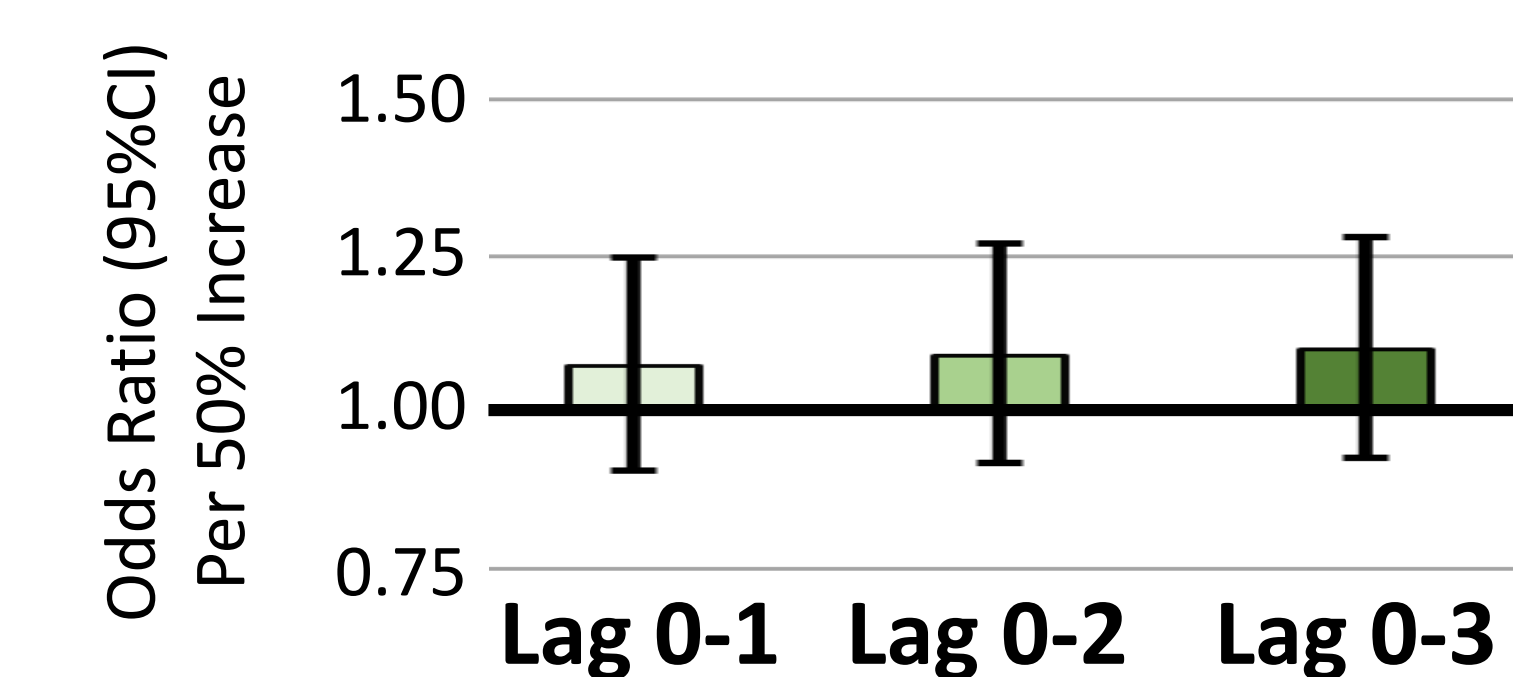


Figure 5: Odds ratio (OR) for having non-shockable rhythm at every 50% increase in average PM2.5 exposure

Number of Days Before Event (Lag)			
	Lag 0-1	Lag 0-2	Lag 0-3
OR	1.07	1.09	1.10
P-value	0.532	0.450	0.408



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